Yuxin Ye

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INDUSTRY EXPERIENCE

Alibaba Group, Tongyi Lab — Algorithm Engineer, RL Infrastructure Mar 2026 – Now

- Parallelize large-scale **ASR / audio-encoder pretraining** for Qwen omni models across datacenter-scale GPU clusters.
- Build GRPO / PPO RL post-training infrastructure for Qwen large-scale **Mixture-of-Experts** models on the verl framework: separated trainer–rollout architecture with **NCCL / NIXL weight synchronization** bridging Megatron training and vLLM inference; reward systems combining rule-based verifiers and LLM-as-judge rubrics.
- Built two-stage multimodal (audio) vLLM-omni rollout — encoder/decoder disaggregation with **shared-memory KV / hidden-state transfer** — wiring trainable-encoder GRPO end-to-end.
- Prototyping MCTS-over-CoT with speculative decoding and per-instance test-time training on this stack (see Research), reusing the GRPO trainer and vLLM-omni rollout as the search and online-update engine.

Cerebras Systems — Member of Technical Staff Jan 2025 – Mar 2026

- Built RL post-training on **wafer-scale hardware (CSX / WSE)**: wrote the trainer API from scratch against its instruction set, memory model, and compiler stack, and stood up a verl-based RL stack — training–inference co-design on non-GPU silicon.
- Deployed and debugged datacenter-scale Cerebras clusters for reliable large-model training / inference.

Lamini AI (enterprise LLM platform; team acquired by AMD, 2025) — HPC Engineer May 2024 – Jan 2025

- Co-authored *Banishing LLM Hallucinations Requires Rethinking Generalization* (Lamini Memory Tuning) with Dr. Gregory Diamos (MLPerf co-founder).
- Enabled **RoCE RDMA** over AMD MI300 / MI250 (Mellanox / Broadcom drivers, Kubernetes / Slurm); implemented DP and FSDP with mpi4py over a 100 Gbps RoCE network; deployed vLLM-compatible serving across NVIDIA and AMD machines.

Amazon Robotics — Software Development Engineer Jul 2022 – Mar 2024

- Developed features for an event-driven, multi-agent simulator of automated sortation floors: multi-robot drive-path planning, staging-cell distribution, resource-based allocation.

PUBLICATION

- Johnny Li, Saksham Consul, Eda Zhou, James Wong, Naila Farooqui, **Yuxin Ye**, Nithyashree Manohar, Zhuxiaona Wei, Tian Wu, Ben Echols, Sharon Zhou, Gregory Diamos. *Banishing LLM Hallucinations Requires Rethinking Generalization.* arXiv:2406.17642, 2024.

EDUCATION

Master of Science, Computational Science and Engineering — Harvard University, GSAS GPA 3.87 2020 – 2022

B.S. in Physics — University of Science and Technology of China (School of the Gifted Young) GPA 3.7 2016 – 2020

SELECTED PROJECTS

Draft Models as Test-Time-Trained Search Policies ongoing

- A single small draft model plays two roles — speculative-decoding proposer verified by a larger target, and rollout policy for MCTS over chain-of-thought — and is *test-time trained* per problem from the search's own outcomes. Load-bearing claim: iso-FLOP accuracy at matched inference compute vs. test-time-scaling and offline-RL baselines.

Gomoku Solver — Monte Carlo Tree Search 2021 – 2022

MIT 6.877 Principles of Autonomy and Decision Making

- UCT MCTS for 9×9 Gomoku with a neural network as the default rollout policy — an AlphaZero-style search + learned-policy agent.

Reinforcement Learning of Soft Fish-like Swimmers (C++) 2021 – 2022

Research Assistant, Prof. Petros Koumoutsakos, Harvard SEAS · github.com/YYXLN/CUP2D

- Coupled fluid–soft-body interaction (inverse-map technique) into a direct numerical flow simulator built on the ACM Gordon Bell Prize-winning Cubism library with Adaptive Mesh Refinement HPC.
- Parametrized the swimmer gait by neural spline knots and trained swimming behavior via reinforcement learning over muscle actuation.

Wasserstein Learning of Generative Models Apr – May 2021

MIT 6.838 Shape Analysis

- Built WGAN-GP from scratch, validated its tractable loss and a Monge-map construction, and benchmarked against W2GAN on MNIST and CIFAR-10.

Realtime Snow Simulation and Rendering (C++) Nov – Dec 2020

MIT 6.837 Computer Graphics · course project contest runner-up

- Simulated snow with the moving-least-squares material point method (MLS-MPM) and rendered in real time.

Auto-differentiation Package (C++) Oct – Dec 2020

github.com/CS107-off-piste/cs107-FinalProject

- Implemented automatic differentiation in both forward and reverse modes, including expression-graph construction and derivative transforms — foundational to compiler / IR and codegen work.

TEACHING

Teaching Assistant — CS 205: Computing Foundations for Computational Science (HPC / HTC), Harvard GSAS Spring 2022

RESEARCH INTERESTS

Reinforcement pretraining (RPT) and the RLVR↔RPT boundary; test-time compute and search-guided self-improvement; encoder compression as the source of predictive representations; inverse RL for alignment; discrete flows / non-autoregressive generation; IO and data movement as a first-class variable in architecture design.

SKILLS

Research: reinforcement learning (GRPO / PPO / DPO), inverse RL, reinforcement pretraining, test-time compute & inference scaling, MCTS, speculative decoding, test-time training, flow matching, reward modeling.

ML / RL Systems: PyTorch, JAX, Triton, Megatron-LM, DeepSpeed, verl, vLLM, CUDA, ROCm, RDMA (RoCE), NCCL / NIXL, MPI / mpi4py, FSDP.

Programming / Infra: C/C++, Python, Go, Julia; Linux, Kubernetes, Slurm, Git, AWS.